"Hope in Every Beat : Revolutionizing Chronic Disease Care"

Remote Patient Monitoring: Continuous Health Tracking, Chronic Disease Management.

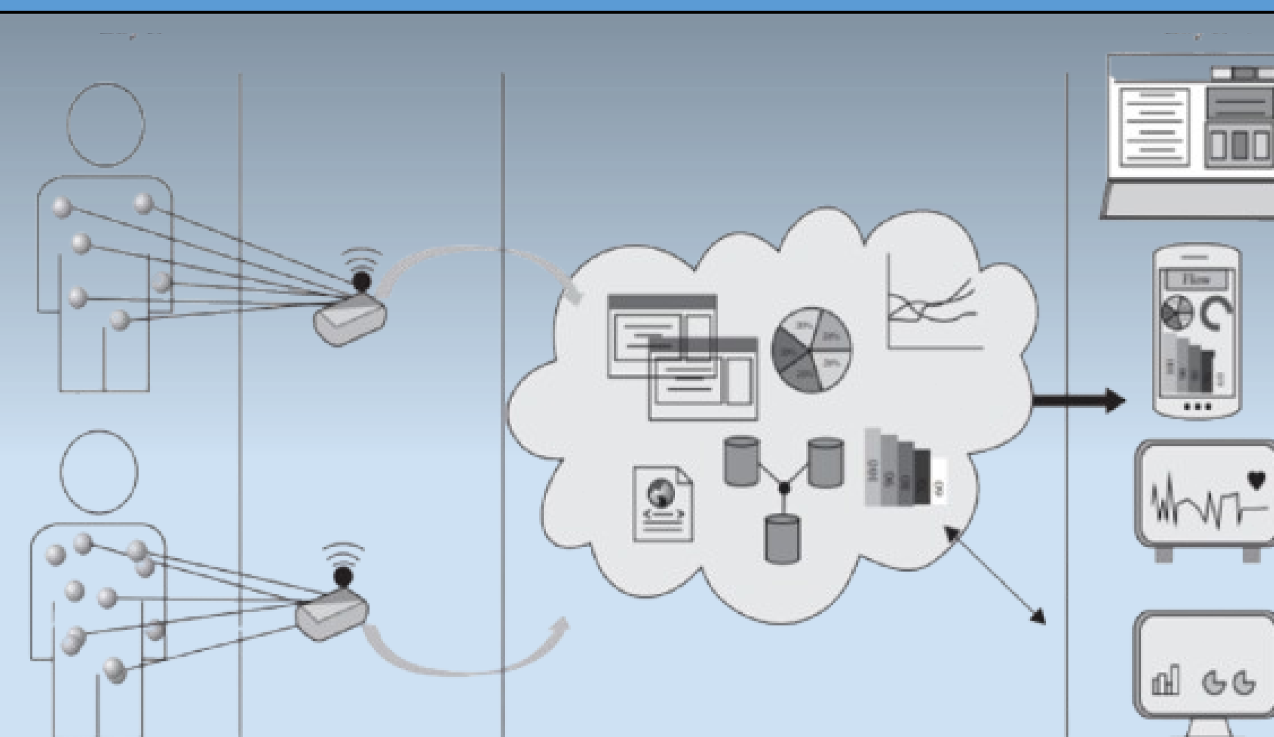
3 Idiots (2009) Raju's Father's Medical Emergency

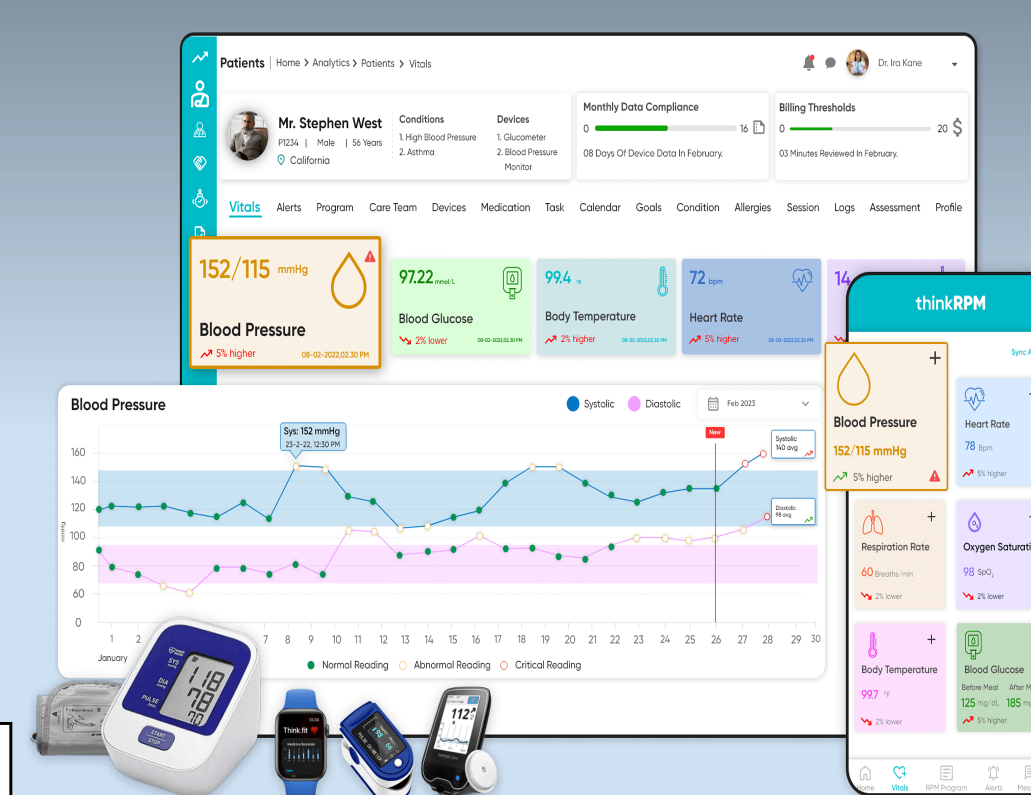
"What if real-time health monitoring could prevent such emergencies? Enter IoT-powered Remote Patient Monitoring."

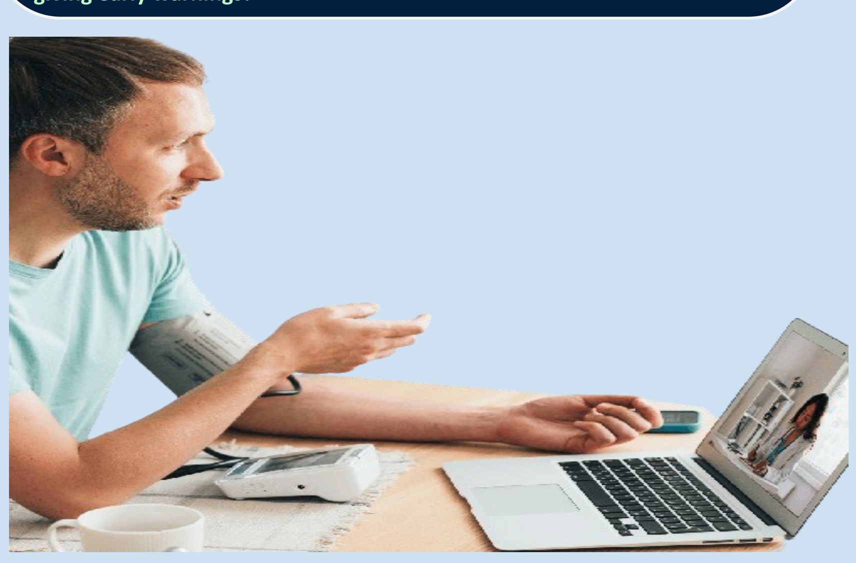
Could RPM Have Helped?

YES! If they had Remote Patient Monitoring (RPM), his vitals—like oxygen levels and heart rate—could have been monitored in real-time. A doctor could have been alerted early, giving instructions immediately instead of waiting until it became a crisis.

So, his condition was treatable, but the delay made it more dangerous. That's exactly why RPM is important—it helps prevent emergencies by giving early warnings!







IoT-based healthcare devices provide access and knowledge about human physiological conditions through hand held devices. With this development, users can be aware of the risks in acquiring various diseases and take necessary precautions to avoid preventable diseases. The basic skeleton of an IoT-based healthcare system is very similar to the conventional IoT architectures. However, for IoT-based healthcare services, the sensors are specifically designed to measure and quantify different physiological conditions of its users/patients.

Medical Sensors & Their Purpose in RPM Devices

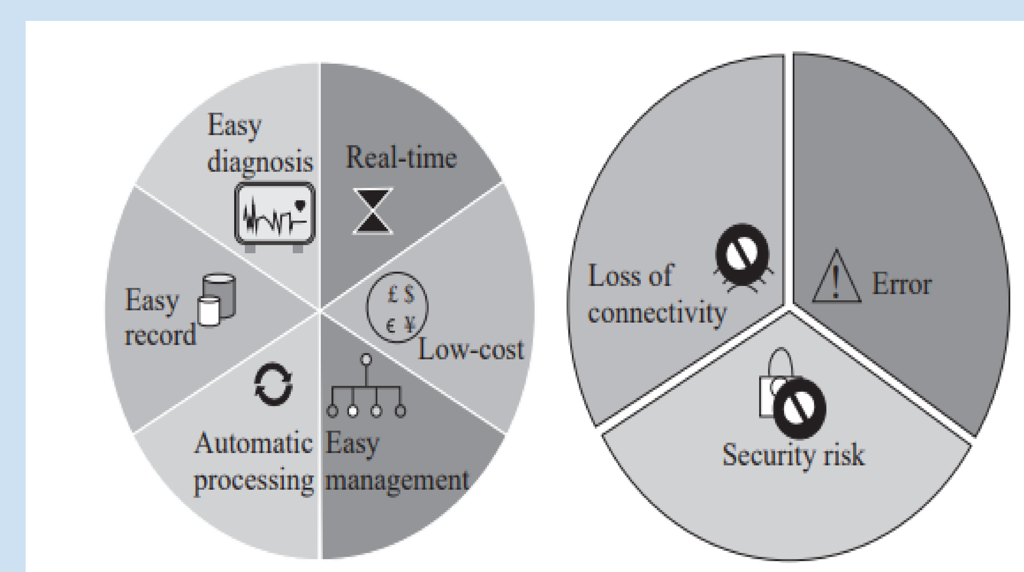
Sensor Type	Function	Common RPM Device
Pulse & Oxygen (SpO2) Sensor	Measures pulse rate and blood oxygen saturation	Pulse oximeter, Smartwatches, Wearable health monitors
Airflow (Breathing Sensor)	Detects respiratory rate and airflow changes	Breathing sensor (as shown in image)
Temperature Sensor	Measures body temperature variations	Continuous Core Temperature Monitor
Blood Pressure Sensor	Measures systolic, diastolic, and mean arterial pressure	Digital Wrist Blood Pressure Monitor
Glucometer Sensor	Tracks blood glucose levels	Blood Glucose Monitor
Galvanic Skin Response (GSR) Sensor	Measures skin conductivity to assess stress levels	GSR Sensor (as shown in image)
Electrocardiogram (ECG) Sensor	Measures electrical activity of the heart	ECG machine, Smart ECG wearables
Electromyogram (EMG) Sensor	Evaluates muscle and nerve function	EMG Monitor

Network Communication Used in These Sensors

Sensor Type	Network Technology Used	Communication Range
Pulse & Oxygen (SpO2) Sensor	Bluetooth (BLE), Wi-Fi, Cellular (4G, LTE-M)	Short-range (BLE), Long-range (Wi-Fi, Cellular)
Airflow (Breathing Sensor)	Bluetooth (BLE), Wi-Fi	Short-range (BLE), Home network (Wi-Fi)
Temperature Sensor	Zigbee, Wi-Fi, Cellular (NB-IoT)	Home network (Zigbee, Wi-Fi), Remote (Cellular)
Blood Pressure Sensor	Bluetooth (BLE), Wi-Fi	Short-range (BLE), Home network (Wi-Fi)
Glucometer Sensor	Bluetooth (BLE), Cellular (NB-IoT, LTE-M)	Short-range (BLE), Long-range (Cellular)
Galvanic Skin Response (GSR) Sensor	Bluetooth (BLE), Wi-Fi	Short-range (BLE)
Electrocardiogram (ECG) Sensor	Bluetooth (BLE), Wi-Fi, Cellular (4G, 5G)	Short-range (BLE), Remote monitoring (Wi-Fi, Cellular)
Electromyogram (EMG) Sensor	Bluetooth (BLE), Wi-Fi	Short-range (BLE), Home network (Wi-Fi)

Communication Protocols Used in These Sensors

Sensor Type	Protocols Used	Purpose
Pulse & Oxygen (SpO2) Sensor	BLE GATT, MQTT, HTTP/HTTPS	Real-time SpO2 data transmission
Airflow (Breathing Sensor)	BLE GATT, MQTT	Sends breathing rate data to cloud or mobile apps
Temperature Sensor	Zigbee Health Profile, MQTT, HTTP/HTTPS	Transfers body temperature readings
Blood Pressure Sensor	BLE GATT, HL7 (EHR Integration)	Syncs BP readings with health records
Glucometer Sensor	BLE GATT, HL7, MQTT	Sends glucose levels to healthcare providers
Galvanic Skin Response (GSR) Sensor	BLE GATT, HTTP/HTTPS	Transfers stress level readings
Electrocardiogram (ECG) Sensor	BLE GATT, DICOM (for ECG Images), HL7	ECG data transmission to EHR
Electromyogram (EMG) Sensor	BLE GATT, MQTT, HTTP/HTTPS	EMG data sharing for muscle function analysis



(a) Advantages of healthcare IoT (b) Risk in healthcare IoT

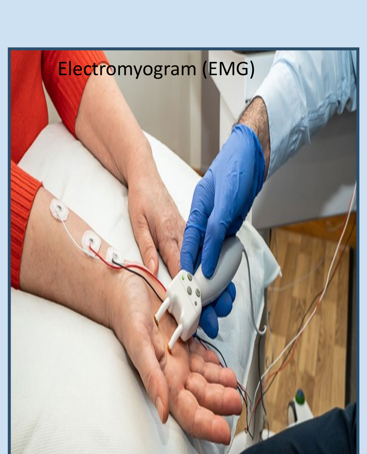
Case Study AmbuSens. ...

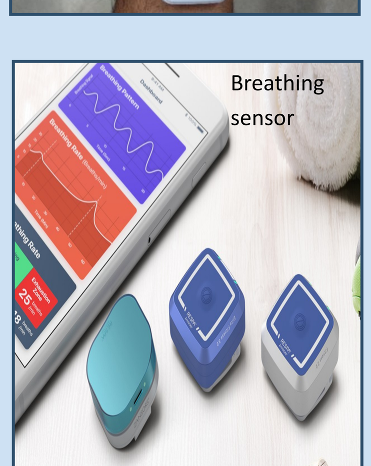
The Future of Healthcare in Big Hero 6 .

Big Hero 6 (2014) – Baymax Personal Healthcare Robot :)

Comparison to 3 Idiots:

Unlike Raju's father, who lacked immediate medical assistance, Baymax's AI-powered IoT system could detect health issues early and provide real-time support, assisting doctors and reducing the risk of emergencies.





Conclusion

Remote Patient Monitoring improves healthcare with real-time tracking, better chronic disease management, and fewer hospital visits, paving the way for smarter, accessible care.

References

- *Big Hero 6* (2014). Walt Disney Animation Studios & *3 Idiots* (2009). Vinod Chopra Films
- Google Images (for visuals).
- ChatGPT (for information).

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